The Postulates of Quantum Mechanics (Cohen–Tannoudji, Ch. III) Section A. Statement of the postulates

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Outline

- A. Statement of the postulates
- B. The physical interpretation of the postulates concerning observables and their measurements
- C. The physical implications of the Schrödinger equation

From classical to quantum description

In classical mechanics, a system is described by

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Quantum mechanics requires a radically different description.

Quantum state space

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This concept can be generalized to *any* physical system.

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Since $\mathcal{E}$ is a vector space, the superposition principle follows naturally: a linear combination of state vectors is a state vector.

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Such an operator is called an *observable*.

Measurement outcomes

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- Since observables are Hermitian, all measurement results are real.
- If the spectrum of A is discrete, the results that can be obtained by measuring A are quantized.

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This prediction is inherently probabilistic.

Discrete non-degenerate spectrum

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Any state can be expanded as

$$|\psi\rangle = \sum_n c_n\,|u_n\rangle\,.$$

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Fourth Postulate:
If the system is in the normalized state $|\psi\rangle$, the probability of obtaining the non-degenerate eigenvalue a_n is

$$\mathcal{P}(a_n) = |\langle u_n|\psi\rangle|^2 = |c_n|^2.$$

where $|u_n\rangle$ is the normalized eigenvector of A associated with the eigenvalue a_n.

Degenerate spectrum

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The probability becomes

$$\mathcal{P}(a_n) = \sum_{i=1}^{g_n} |\left\langle u_n^{(i)}\middle|\psi\right\rangle|^2 = \sum_{i=1}^{g_n} |c_n^{(i)}|^2.$$

Projectors and basis independence

Define the projector onto the eigensubspace $\mathcal{E}_n$:

$$P_n = \sum_{i=1}^{g_n} \left|u_n^{(i)}\right\rangle \left\langle u_n^{(i)}\right| . \quad |\psi_n\rangle = P_n |\psi\rangle = \sum_{i=1}^{g_n} c_n^i \left|u_n^{(i)}\right\rangle$$

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The probability $\mathcal{P}(a_n)$ can be written as

$$\mathcal{P}(a_n) = \sum_{i=1}^{g_n} |c_n^{(i)}|^2 = \langle\psi_n|\psi_n\rangle = \langle\psi| P_n^\dagger P_n |\psi\rangle = \langle\psi| P_n |\psi\rangle .$$

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This expression is independent of the chosen basis in $\mathcal{E}_n$.

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The eigenkets form a continuous basis:

$$|\psi\rangle = \int d\alpha\; c(\alpha)\,|v_\alpha\rangle\,.$$

Probability density

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with

$$\rho(\alpha) = |c(\alpha)|^2 = |\, \langle v_\alpha | \psi \rangle \,|^2.$$

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If the state is not normalized, probabilities must be renormalized accordingly.

Global phase

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$$|\left\langle u_n^{(i)} \middle| \psi' \right\rangle|^2 = |e^{i\theta} \left\langle u_n^{(i)} \middle| \psi \right\rangle|^2 = |\left\langle u_n^{(i)} \middle| \psi \right\rangle|^2.$$

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Therefore, two proportional kets represent the same physical state, as the physical results remain unchanged.

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Global phases are irrelevant (do not affect physical predictions); relative phases of the coefficients of an expansion are significant.

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If the measurement of A yields the simple eigenvalue a_n of A, we postulate that the state immediately after the measurement is the eigenvector $|u_n\rangle$ associated with a_n.

Reduction of the wave packet

If the eigenvalue a_n is degenerate, ψ before the measurement is written:

$$|\psi\rangle = \sum_n \sum_{i=1}^{g_n} c_n^{(i)} \left|u_n^{(i)}\right\rangle$$

Reduction of the wave packet

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Immediately after the measurement is:

$$|\psi\rangle \to \frac{1}{\sqrt{\sum_{i=1}^{g_n} \left|c_n^{(i)}\right|^2}} \sum_{i=1}^{g_n} c_n^{(i)} \left|u_n^{(i)}\right\rangle$$

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If the measurement of $\mathcal{A}$ yields the eigenvalue a_n, the state immediately after the measurement is

$$|\psi\rangle \longrightarrow \frac{P_n |\psi\rangle}{\sqrt{\langle\psi| P_n |\psi\rangle}}.$$

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The system is already in the corresponding eigensubspace. It is the part of $|\psi\rangle$ which belongs to $\mathcal{E}_\backslash$

Time evolution of a quantum system

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The time evolution of the state vector $|\psi(t)\rangle$ is governed by the Schrödinger equation:

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Here $H(t)$ is the observable associated with the total energy of the system.

The Hamiltonian operator

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The Hamiltonian plays a central role in quantum dynamics.

Quantization rules

Goal: given a classical physical quantity $\mathcal{A}(r, p, t)$, construct the corresponding operator A.

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The operator is obtained as

$$A(t) = \mathcal{A}(R, P, t).$$

Canonical commutation relations

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These relations define the quantum structure of phase space.

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In quantum mechanics:

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and neither operator is Hermitian.
The correct observable is obtained by symmetrization:

$$\frac{1}{2}(R \cdot P + P \cdot R).$$

Particle in a scalar potential

Classical Hamiltonian:

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No symmetrization is needed in this case.

Schrödinger equation

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$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \left[\frac{P^2}{2m} + V(R) \right] |\psi(t)\rangle \, .$$